

Challenges (3): data reporting and sharing standards

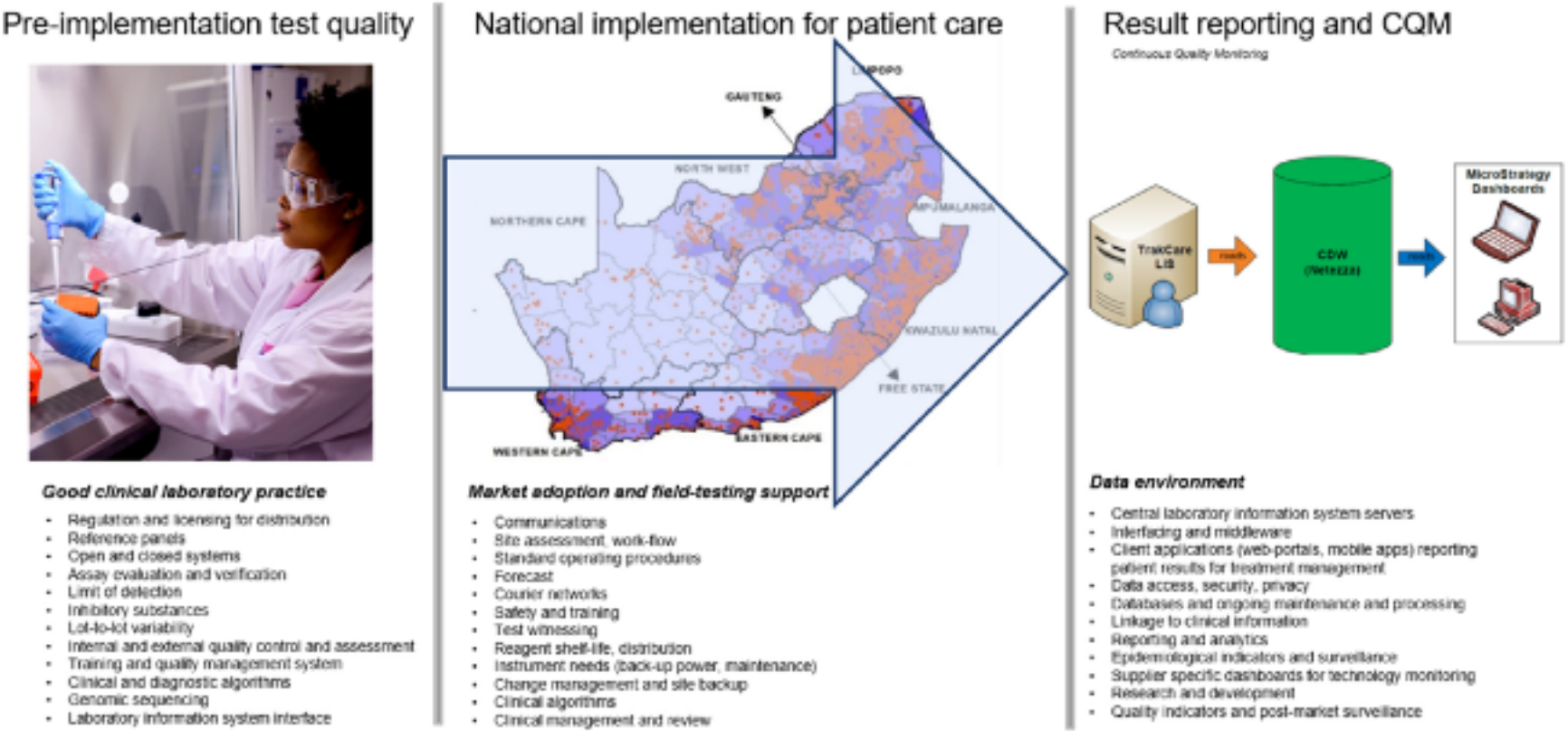
Establishing standards for robust and ethical data reporting/sharing

Nationwide Harmonization Effort for Semi-Quantitative Reporting of SARS-CoV-2 PCR Test Results in Belgium

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How South Africa Used National Cycle Threshold (Ct) Values to Continuously Monitor SARS-CoV-2 Laboratory Test Quality

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Table 1. Proposed interpretation and reporting scheme for SARS-CoV-2 PCR results [24]: a semi-quantitative reporting approach of PCR test results into four categories of SARS-CoV-2 positivity.

Category of SARS-CoV-2 Positivity	SARS-CoV-2 Viral Load (RNA Copies/mL)	Interpretation with Respect to Infectivity
Very strongly positive	$\geq 10^7$	patient is contagious
Strongly positive	$\geq 10^5 - < 10^7$	patient is probably contagious
Moderate positive	$\geq 10^3 - < 10^5$	patient is potentially contagious, unless there is clinical and/or serological evidence of an old, cleared infection
Weakly positive	$< 10^3$	patient is probably not or no longer contagious if there is also clinical and/or serological evidence of an old, cleared infection

Challenges (4): modeling frameworks

Some platforms exist(ed...)



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Viral Kinetic Model

Viral dynamics models provide valuable insight into the life cycle of infectious agents. This in silico viral kinetic model reveals the challenges of monotherapy treatment: a monotherapy treatment must be initiated very early in viral infection and be quite potent. However, combination therapy where different steps of the SARS-CoV-2 life cycle are modestly impacted may be an alternate strategy. Numerous model structures have been proposed, but the model for extremely fast and short-duration replication (e.g., influenza) derive from Baccam 2006. The model used here is from Gonçalves 2020 and expands the infected cell population.

We performed simulations with this viral kinetic model and found that combinations of therapeutics targeting specific rate constants have greater probability of efficacy, especially in the treatment of early infection phase in COVID-19 patients. By targeting multiple points central to viral replication within infected host cells or release from those cells, we may be able to reduce both viral load and host cell infection. Access the full manuscript which is available as a preprint: Michael Dodds, Rajesh Krishna, Antonio Goncalves, Craig Rayner. Model-Informed Drug Repurposing: Viral Kinetic Modeling to Prioritize Rational Drug Combinations for COVID-19. Authorea. June 12, 2020. DOI: 10.22541/au.159200535.51965254Hi

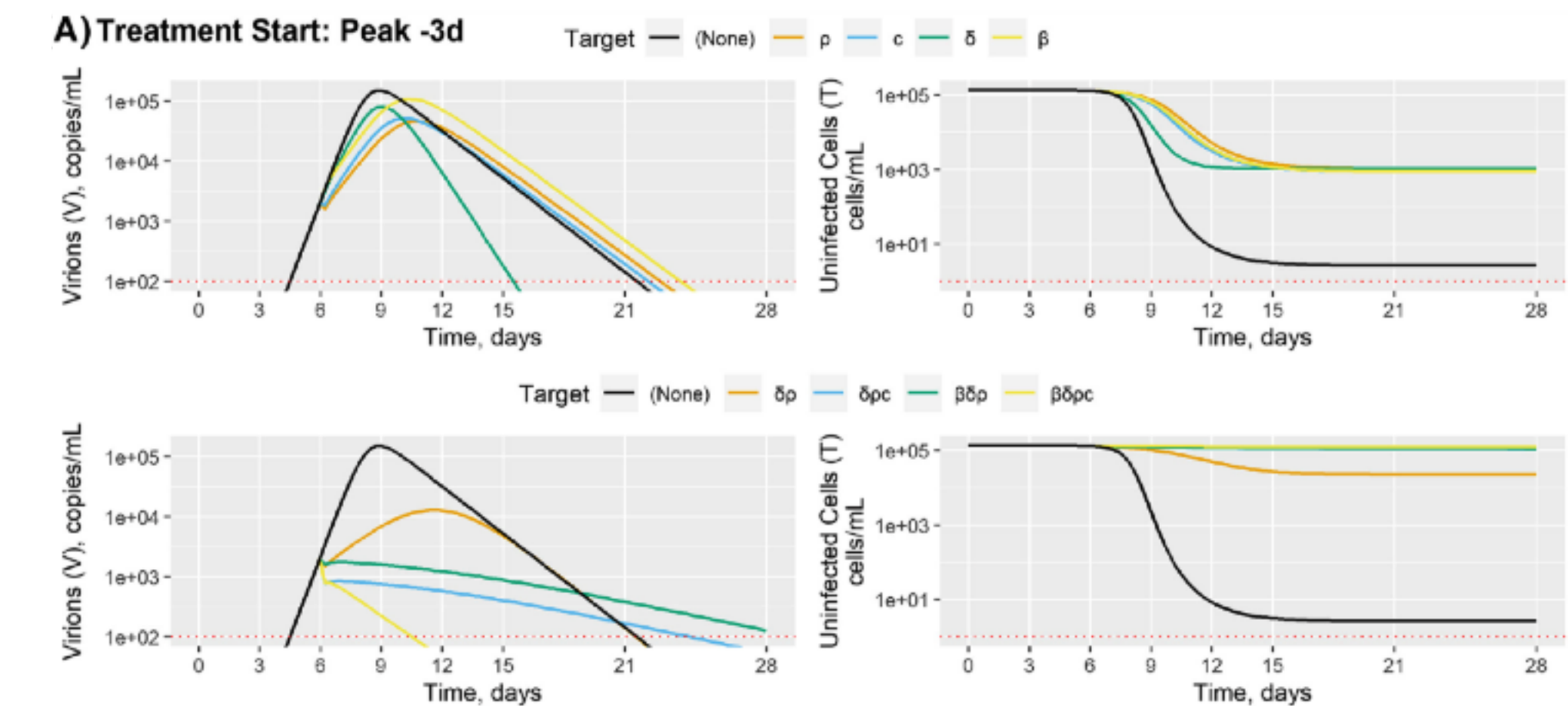


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Model-informed drug repurposing: Viral kinetic modelling to prioritize rational drug combinations for COVID-19

Michael G. Dodds, Rajesh Krishna, Antonio Goncalves, Craig R. Rayner

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Rt estimation a useful paradigm

